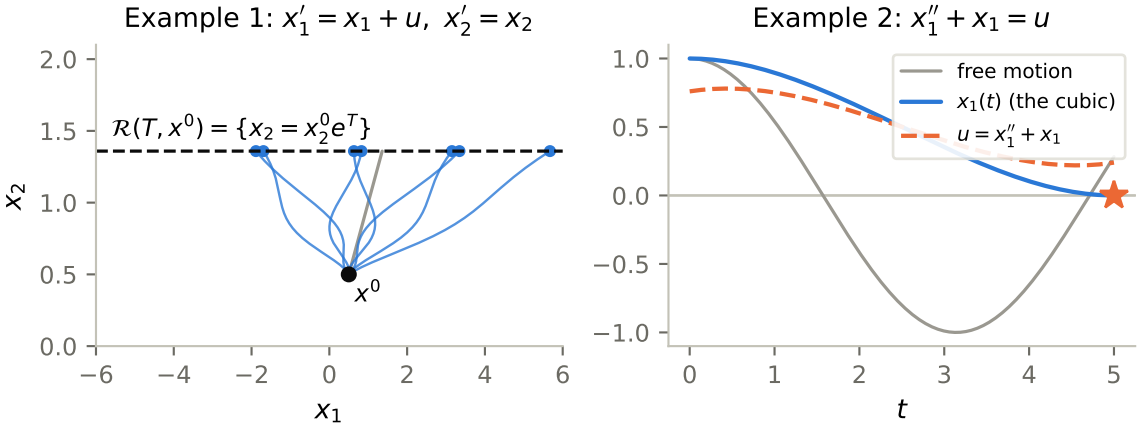
Example 1: $x_1' = x_1 + u,\ x_2' = x_2$
$\mathcal{R}(T, x^0) = \{x_2 = x_2^0 e^T\}$
x^0
x_1
x_2
Example 2: $x_1'' + x_1 = u$
free motion
$x_1(t)$ (the cubic)
$u = x_1'' + x_1$
t